



Alveolar cleft reconstruction utilizing a particulate autogenous tooth graft and a novel split-thickness papilla curtain flap — A retrospective study

Bálint Molnár^{a,1}, Tamás Würsching^{b,c,1,*}, Eleonóra Sólyom^a, Laura Pálvölgyi^b, Zsombor Radóczy-Drajkó^a, Dániel Palkovics^a, Krisztián Nagy^{b,d,e}

^a Department of Periodontology, Semmelweis University, Budapest, Hungary

^b Centre for Facial Reconstruction, Department of Pediatrics, Semmelweis University Budapest, Hungary

^c Department of Oro-Maxillofacial Surgery and Stomatology, Semmelweis University, Budapest, Hungary

^d Cleft and Craniofacial Centre, Division of Maxillo-Facial Surgery, General Hospital St. Jan, Bruges, Belgium

^e OMFS-IMPACT Research Group, Faculty of Medicine, Department of Imaging AndPathology, KU Leuven and Oral and Maxillofacial Surgery, University Hospitals Leuven, Leuven, Belgium

ARTICLE INFO

Handling Editor: Prof. Emeka Nkenke

Keywords:

Secondary alveolar bone grafting
Autogenous tooth bone graft
3D surgical planning
Volumetric radiographic evaluation
Unilateral cleft lip and palate

ABSTRACT

During secondary alveolar cleft grafting, the use of autogenous cancellous bone harvested from the iliac crest is still considered the gold standard. Due to the risk of donor-site morbidity and excessive graft resorption, alternative grafting materials (e.g. intraoral bone, xenografts) have been tested. Autogenous tooth bone graft (ATB) is a novel material derived from extracted teeth. ATB has successfully been used in pre-prosthetic and periodontal surgery for hard-tissue reconstruction.

Seven patients with unilateral cleft lip and palate were treated with ATB, using their own deciduous teeth for grafting. Defects were accessed utilizing a novel split-thickness papilla curtain flap. Cone-beam computed tomography scans were taken prior to and 3 months following cleft surgery to assess graft integration, graft stability, and the volume of the newly formed hard tissues. Hard-tissue gain, as measured at the 3-month follow-up, averaged $0.65 \text{ cm}^3 \pm 0.26 \text{ cm}^3$. Results showed acceptable graft integration and stability at the 3-month follow-up, with no adverse effects or excessive resorption of the graft.

The use of ATB might be a feasible alternative for alveolar cleft grafting. However, long-term studies using a large sample size are required to derive further conclusions.

1. Introduction

The comprehensive treatment of patients with cleft lip and palate is a complex and demanding task that requires the collaborative effort of different medical specialties. One of the most crucial steps in the effective treatment of these patients is alveolar bone grafting, which is necessary to reconstruct the bony defect of the alveolar process, maintain a proper arch form, support the alar base, and close the oronasal fistulae (Abyholm et al., 1981). Timing of the surgery differs across various cleft centers with secondary alveolar bone grafting (SABG) executed most commonly during the mixed dentition (Janssen et al., 2014). The most favorable outcomes have been achieved when the surgery was concluded before the eruption of the cleft-side canine (Brudnicki et al., 2017; Vandersluis et al., 2020).

In terms of local flap design for alveolar cleft closure, the most commonly utilized method is the four-flap technique, as described by Nordin and Abyholm (Eskeland et al., 1979; Larson et al., 1983), which provides an ideal soft-tissue closure above the graft. However, in cases of suture loosening due to flap tension, secondary wound healing might occur, and the consequent exposure might lead to the loss of the graft. This classic flap design does not address the scarce width and thickness of keratinized gingiva at teeth adjacent to the alveolar cleft, where consequent periodontal impairment may develop, resulting in eventual tooth loss (Huynh-Ba et al., 2009; Sahni et al., 2022).

Currently, the most common choice of material for alveolar bone grafting is autologous bone, which is considered to be the gold standard, combining all properties required in a bone graft: osteoconduction, osteoinduction, and osteogenesis. The most preferred donor site is the

* Corresponding author. Bokay Janos Street 53–54, 1083, Budapest, Hungary.

E-mail address: wursching.tamas@semmelweis.hu (T. Würsching).

¹ These authors contributed equally to the preparation of the manuscript.

iliac crest, as described by Boyne and Sands (Boyne and Sands, 1972). However, donor-site morbidity (as high as 19.37%) and excessive graft resorption need to be taken into account (Dimitriou et al., 2011; Eufinger and Leppänen, 2000).

To reduce donor-site morbidity and to possibly increase the success rate of the surgical intervention, various authors have suggested alternative grafting materials for the reconstruction of alveolar clefts (Ishii et al., 2002; Yuan et al., 2004; Alonso et al., 2010; Aly and Hammouda, 2016; Blume et al., 2019; Ellapakurthi and Reddy, 2021; Kadry et al., 2021). Aside from graft materials, the added benefits of supplementary materials, such as fibrin sealant, platelet-rich plasma, and leukocyte-/platelet-rich fibrin have also been evaluated (Wu et al., 2018; Lavagen et al., 2021).

As an alternative to autogenous bone grafts, Kim et al. introduced a method for producing autogenous tooth bone grafts (ATB) by grinding and chemically processing previously extracted teeth (Kim et al., 2017). Due to its microscopic structure, ATB has both osteoconductive and osteoinductive properties, making it a suitable material for alveolar ridge preservation, alveolar ridge augmentation, and alveolar cleft grafting, with successful outcomes (Hara et al., 2013; Jeong et al., 2015; Datarkar and Bhawalkar, 2020; Radoczy-Drajko et al., 2021). Jeong et al. were able to place a 3 mm-wide implant with 45 Ncm insertion torque 3.5 months after cleft grafting (Jeong et al., 2015).

For adequate treatment planning, it is necessary to identify the extent and the shape of the alveolar cleft. In recent years, cone-beam computed tomography (CBCT) has become widespread as an imaging modality in dentistry and maxillofacial surgery (Jacobs et al., 2018). CBCT enables 3D visualization of the alveolar bone defect morphology, digital planning, and additive manufacturing of bone grafting templates. These processes aid the surgeons during the intervention, and may reduce further complications (Pálházi et al., 2014). CBCT scans can also be utilized for postoperative assessment of graft integration and volume stability (Linderup et al., 2015; Blume et al., 2019; Stasiak et al., 2019), since conventional methods lack the possibility of volumetric assessment and tend to over- or underestimate the size of the graft (Palankar et al., 2021; Kamperos et al., 2020).

To overcome the limitations of current methods utilizing advancements provided by novel treatment approaches, the aim of our pilot case series was to evaluate the efficacy of a digitally planned ATB powder block — stabilized with fibrin sealant — combined with a novel split-thickness papilla curtain flap in the treatment of unilateral alveolar cleft defects.

2. Materials and methods

2.1. Study design and patient selection

Patients exhibiting unilateral cleft lip and palate (UCLP) were recruited for this study. Enrolled patients were under treatment in the Department of Pediatrics, Semmelweis University, Budapest. The study was approved by the university's Regional and Institutional Committee of Science and Research Ethics (approval number: SE RKEB 251/2020). The study protocol was submitted and approved by the US National Library of Medicine (www.clinicaltrials.gov; trial registration number: NCT05971914). The research plan was compiled in line with current legislation and the Declaration of Helsinki of the World Medical Association (reference number: 23/2002. V.9.). Surgical interventions were undertaken with the understanding and written consent of each subject's caregiver. The exclusion criteria were: (i) major relevant general medical conditions; (ii) systemic use of steroids; (iii) current or previous intravenous bisphosphonate treatment; (iv) acute infection at the operation site; and (v) previous attempts at alveolar bone grafting. In every case the development of the cleft-side canine was considered during planning of the surgery, as was the need for orthodontic treatment. All the patients included in the study required orthodontic maxillary expansion. The development of the canine was followed using

periapical X-rays. Only once the canine's root development was between one-third and two-thirds of its expected final length, was the decision made to proceed with surgery. At this point CBCT scans were taken for surgical planning. Each patient were required to have presented at least three deciduous teeth, which were scheduled for extraction. All enrolled patients were examined by two independent dentists to assure that tooth removal was indicated according to patients' age and orthodontic planning.

2.2. Radiographic image acquisition and image processing

CBCT scans were acquired with a 12 cm × 8 cm field of view, and a 200 µm voxel size (VGI EVO®, Newtom). CBCT data were acquired at baseline and at the 3-month follow-up. At the same time points, the width of keratinized gingiva at the adjacent teeth and over the alveolar cleft was recorded using a UNC-15 periodontal probe.

Digital imaging and communications in medicine (DICOM) datasets were imported into 3D Slicer open-source image-processing software (www.slicer.org) (Pinter et al., 2019). Baseline and follow-up CBCT scans were segmented utilizing a semi-automatic segmentation method in the 'Segment Editor' module of 3D Slicer (Palkovics et al., 2021b). Separate binary label maps were generated for the bone of the maxilla. The maxilla was segmented utilizing the region-growing tool 'Grow from seeds', and thereafter the 'Watershed' tool was used to segment all the teeth separately (Pinter et al., 2019). The software automatically generated 3D models of binary label maps, with real-time rendering (Fig. 1).

2.3. The virtual surgical planning process

Based on the extent and morphology of the cleft the desired shape of the nasoalveolar graft was designed in 3D Slicer. Since every case was unilateral, the non-cleft side anatomy was used as a reference during graft planning. This also gave us the opportunity to determine the exact volume of the graft in cm³. This was a helpful addition in planning the surgery because the maximum amount of graft the ATB machine can produce is approximately 3 cm³. Another limiting factor is the number and condition of removable deciduous teeth.

After planning, the model was exported as a standard tessellation language (STL) file. As the final step, a grafting template was constructed using an open-source computer-aided design (CAD) software (Blender®; Blender Foundation, Amsterdam, The Netherlands), as described by Fabian et al. (2019) (Fig. 2). The designed parts were manufactured with a stereolithography (SLA) 3D printer (Phrozen Shuffle XL; Phrozen, Hsinchu City, Taiwan) using class I biocompatible, surgical-grade resin (Dental SG; NextDent BV, Soesterberg, The Netherlands).

2.4. Surgical protocol

Teeth were extracted under general anesthesia with the gentle application of forceps and elevators. Extracted teeth were prepared immediately after removal, according to the Bonmaker® protocol (Korea DentalSolution Co Ltd, Busan, Korea) (Kim et al., 2017) (Fig. 3). The outer surface of the teeth was debrided with diamond-coated burs. Following the removal of pulp tissues, carious lesions, root canal fillings, and restorations, the crowns and roots were both crushed in the Bonmaker® tooth grinder. After grinding, the tooth particles (particle size: 425–1500 µm) underwent disinfection and preparation via proprietary A, B, and C liquids inside the Bonmaker® device (Kim et al., 2017). Between patients, the equipment underwent a regular disinfection cycle, according to the manufacturer's instructions. During the preparation and disinfection procedure (which usually takes 30–35 min), flap preparation and exploration of the alveolar cleft and oronasal fistula were carried out.

Articaine hydrochloride 4% with 0.0001% epinephrine (Ultracain DS

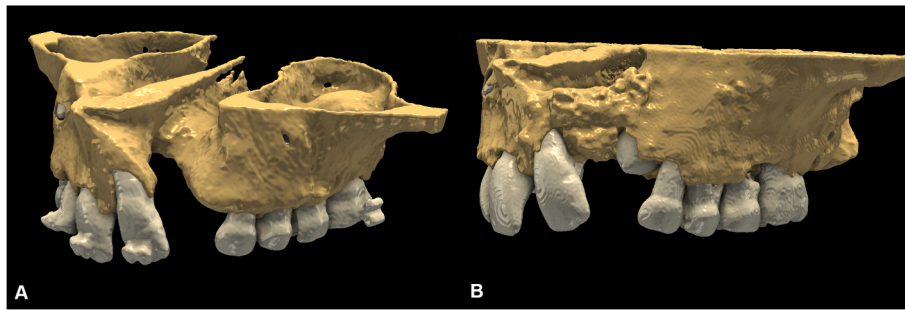


Fig. 1. Segmented 3D models of CBCT datasets: (A) virtual model of baseline hard-tissue conditions; (B) virtual model of 3-month follow-up hard-tissue conditions.

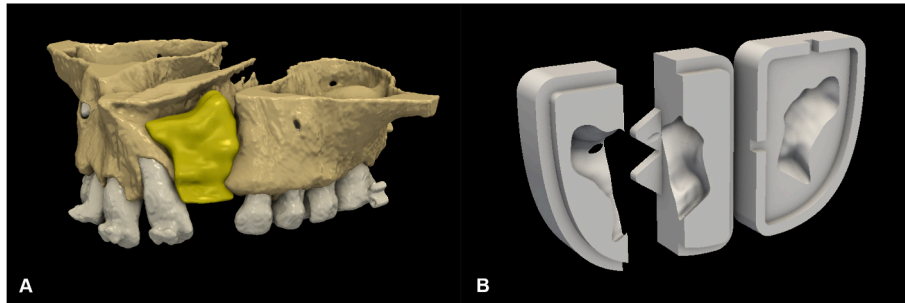


Fig. 2. Preoperative virtual planning: (A) digital 3D plan of cleft reconstruction; (B) CAD-modelled grafting template.

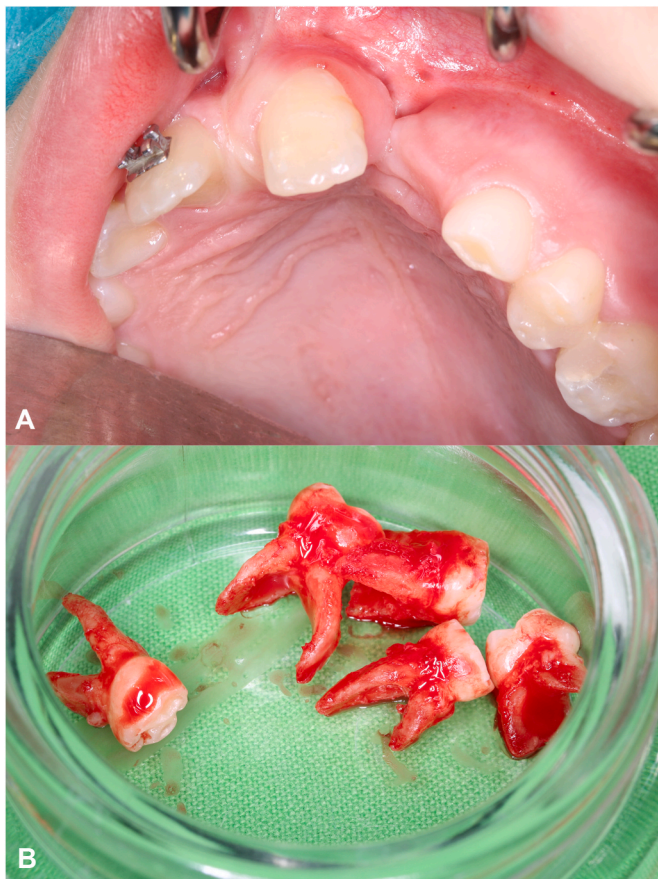


Fig. 3. Baseline clinical situation and tooth extraction: (A) baseline clinical situation; (B) extracted deciduous teeth.

Forté; Sanofi-Aventis, Paris, France) was used for local anesthesia. A full-thickness incision was made starting at the ipsilateral maxillary tuberosity, continued in a continuous intrasulcular incision along all teeth located distally from the oronasal fistula, with a no. 15 blade (Aesculap; Braun AG, Tuttlingen, Germany). At the level of the cleft and the oronasal fistula (located in the lateral incisor position), the incision was continued along the alveolar crest in full-thickness, followed by a split-thickness incision at the mucogingival junction of the adjacent central incisor, and continued vertically in the midline labial frenulum. Subsequently, the area of keratinized gingiva around teeth located distally from the cleft was elevated in full-thickness with blunt dissection. Flap elevation was continued by sharp dissection apically from the mucogingival junction of the distal teeth, as well as at the loose mucosa over the central incisor, where the keratinized gingiva was left untouched. The periosteal layer of the complete hemimaxilla was then exposed, followed by excision of the epithelial down growth in the oronasal fistula, which was closed with resorbable monofilament sutures (Chirmax Monolac 5/0) by approximating adjacent scar tissue margins. Scar tissue in the cleft area was outlined by mesial, distal, and palatal incisions to allow for nasal repositioning, exposing the bony cleft and creating a secluded space for graft insertion (Figs. 4 and 5).

Ready-to-use ATB was mixed with fibrin sealant (Segura-Castillo et al., 2005) (Tisseel®; Baxter, Glenview, Illinois, USA) in the digitally planned and 3D-printed grafting templates to obtain a sticky ATB graft that matched the shape and extent of the bony defect. The pre-shaped sticky ATB graft was then inserted in the cleft defect (Fig. 6).

Following cleft augmentation, the split-thickness papilla curtain flap was repositioned by shifting all buccal surgical papillae mesially to the adjacent or the second-adjacent interproximal space, depending on the horizontal extent of the cleft. Next, the papilla in the close vicinity of the oronasal fistula was relocated to the level of the mesial vertical incision in the midline frenulum (Fig. 7). As a result, healthy continuous keratinized gingiva was shifted over the cleft to cover the grafted area and to ensure simultaneous reconstruction of the distorted vestibulum and lacking keratinized tissues. To stabilize the tension-free split-thickness papilla curtain flap, suturing was carried out with resorbable monofilament sutures (Chirmax Monolac 5/0), utilizing circumferential double

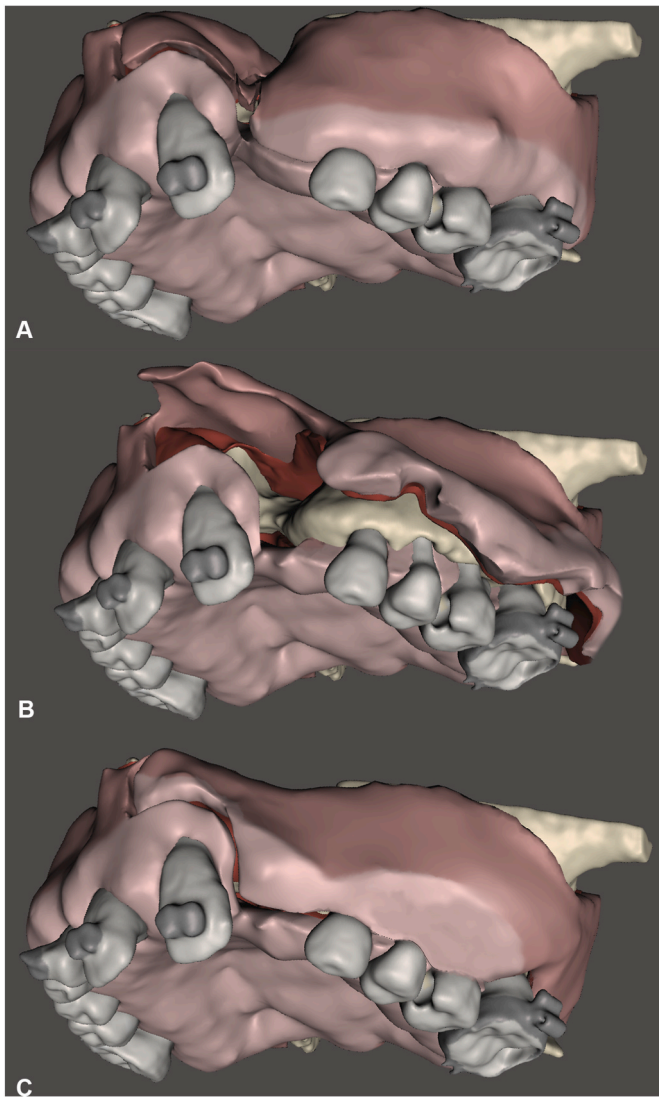


Fig. 4. Schematic presentation of the split-thickness papilla curtain flap: (A) initial incision (light pink: keratinized mucosa, dark pink: mobile mucosa); (B) flap elevation (red: periosteum); (C) mesially shifted flap closure.

sliding sutures and single interrupted mucosal sutures (Fig. 8A).

Postoperative systemic antibiotic therapy (875/125 mg amoxicillin/clavulanic acid twice a day, or 300 mg clindamycin four times a day in cases of penicillin allergy) was prescribed. Patients were advised to clean the area gently with an ultra-soft toothbrush and rinse with a 0.2% chlorhexidine-digluconate mouthwash for 2 weeks after surgery. Sutures usually resorbed 3–4 weeks after surgery (Fig. 8B).

2.5. Three-dimensional radiographic evaluation

Following segmentation of both baseline and 3-month follow-up CBCT scans, spatial registration was carried out to validate volumetric and morphological hard-tissue alterations. Registration of DICOM datasets was performed using the ‘General Registration (Elastix)’ module, which is an automatic intensity-based medical image registration algorithm (Klein et al., 2010). After spatial registration, the preoperative 3D model of the maxilla was subtracted from the postoperative 3D model (Palkovics et al., 2021a, 2022a, 2022b). Subsequently, the hard-tissue difference between the two time points could be visualized and evaluated in 3D. The volumes of the newly formed hard tissues were calculated and compared with the preoperatively planned volumes.

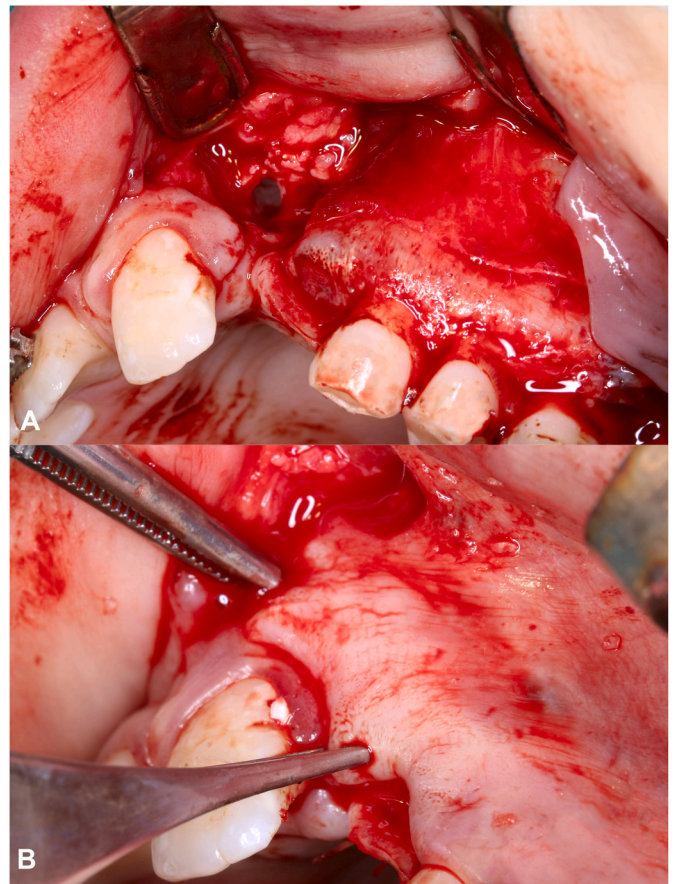


Fig. 5. Intraoperative situation: (A) surgical access of the UCLP; (B) tension-free mesial flap advancement.

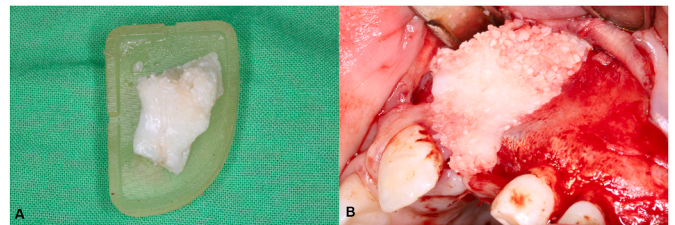


Fig. 6. Hard-tissue reconstruction of the UCLP: (A) the mixture of fibrin sealant and ATB particles was shaped by the 3D-printed grafting template to prepare the sticky ATB graft; (B) reconstruction of the UCLP with the sticky ATB graft.

2.6. Outcome measures

The primary outcome of the current investigation was the evaluation of volumetric hard-tissue gain by 3D subtraction analysis.

The secondary outcomes of the study were: (i) validation of the efficacy of the surgical procedure and volumetric stability of the graft, by calculating the ratio between the actual volumetric hard-tissue gain and the planned hard-tissue volume; and (ii) assessment of the similarity between the planned and actual hard-tissue morphologies by calculating the Dice similarity coefficient (DSC) and Hausdorff distance (HD). DSC is a spatial overlap index and a reproducibility validation metric. The value of DSC ranges from 0, indicating no spatial overlap between two sets of binary segmentation results, to 1, indicating complete overlap (Janssen et al., 2017). HD is the longest distance from a point in one of the two models to its closest point in the other model, meaning that the lower the HD value the more the two models overlap.

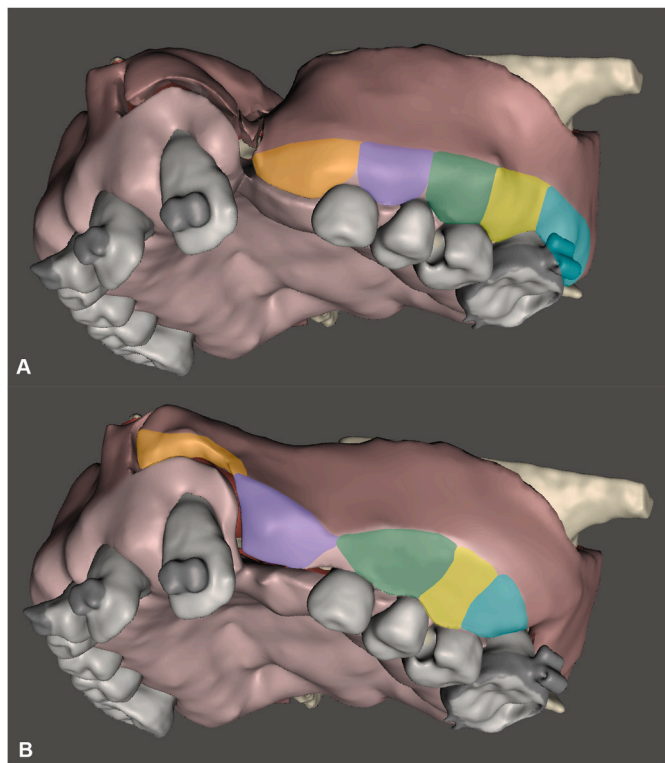


Fig. 7. Schematic demonstration of the mesial papilla shift: (A) preoperative state (the first interdental papilla is located distally to the alveolar cleft, marked in purple); (B) Postoperative state (the interdental papilla, marked in purple, was shifted mesially to cover the area of the alveolar cleft).

2.7. Statistical analysis

The primary focus of the study was to determine the amount of hard tissue change rather than determining absolute baseline and follow-up values. Therefore, only descriptive statistics were used. Data were expressed as mean value $\pm$ standard deviation.

3. Results

3.1. Patient demographics

In total, seven participants exhibiting UCLP were enrolled in this investigation. Five patients were male, two were female. The patients were aged 9–11 years, with a mean age of 10.43 ± 0.79 years. Two defects were located on the left side and five on the right. In total, 35 deciduous teeth and two supernumerary teeth were extracted and utilized for preparation of the sticky ATB graft. In every case, the patient's own teeth provided enough material for proper reconstruction of the alveolar defect, using an average of 4.75 ± 2.49 teeth per patient. Patient demographic data are shown in Table 1.

3.2. Clinical outcomes

All patients were able to be discharged from the hospital on the first postoperative day without any major complaints, and to return to normal daily activity shortly after surgery. Initial wound healing was uneventful in all seven cases, with no signs of graft exposure over the postoperative period. The quality of keratinized tissues was favorable, fully eliminating any muscle pull or vestibular distortions over the cleft. During the follow-up period, eruption of the cleft-side canine was observed in one patient.

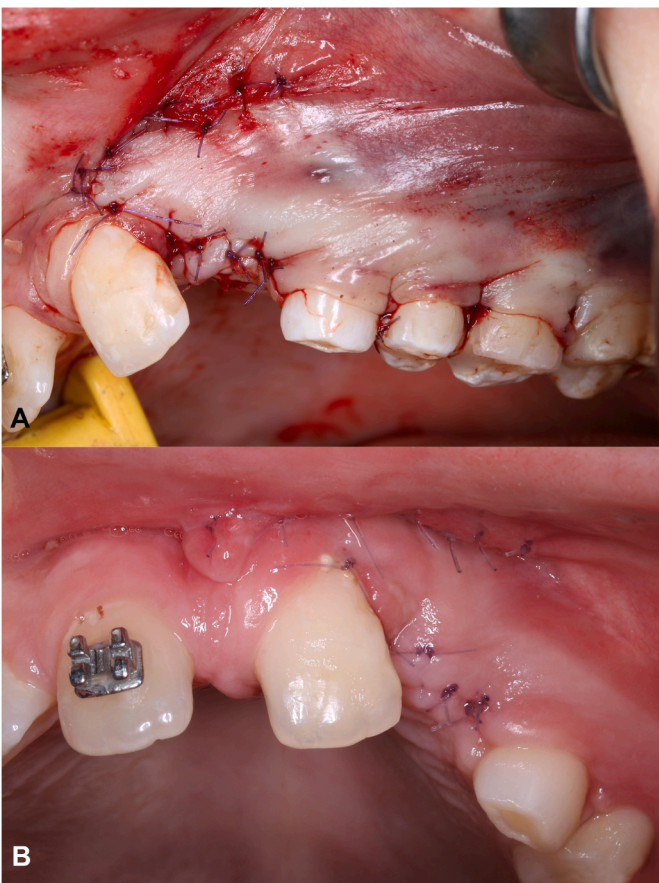


Fig. 8. Wound closure and postoperative control: (A) wound closure with primary intention; (B) 2-week postoperative control.

Table 1
Patient demographic and surgical data.

Patient	Age	Sex	No. Of extracted teeth	Surgical area location
1	11	M	4	Right side
2	10	M	5	Right side
3	11	M	10	Left side
4	11	F	4	Left side
5	10	M	4	Left side
6	9	M	5	Left side
7	11	F	5	Left side
Mean $\pm$ standard deviation	10.43 ± 0.79	n. a.	5.29 ± 2.14	n.a.

3.3. Primary outcome measure — volumetric hard tissue alterations

Planned graft volumes averaged $1.14 \text{ cm}^3 \pm 0.36 \text{ cm}^3$, whereas hard-tissue gain, as measured on the 3-months follow-up CBCT scans, averaged $0.65 \text{ cm}^3 \pm 0.26 \text{ cm}^3$ (Fig. 9). The primary outcome data are summarized in Table 2.

3.4. Secondary outcome measures

The ratio between actual and planned graft volume ranged between 34.17% and 108.21%, resulting in an average ratio of $59.92\% \pm 24.35\%$. DSC between planned and actual graft segmentation ranged between 0.14 and 0.65, with an average of 0.43 ± 0.2 for all seven cases. HD averaged $1.83 \text{ mm} \pm 0.77 \text{ mm}$ (Fig. 10). Secondary outcome data are summarized in Table 3.

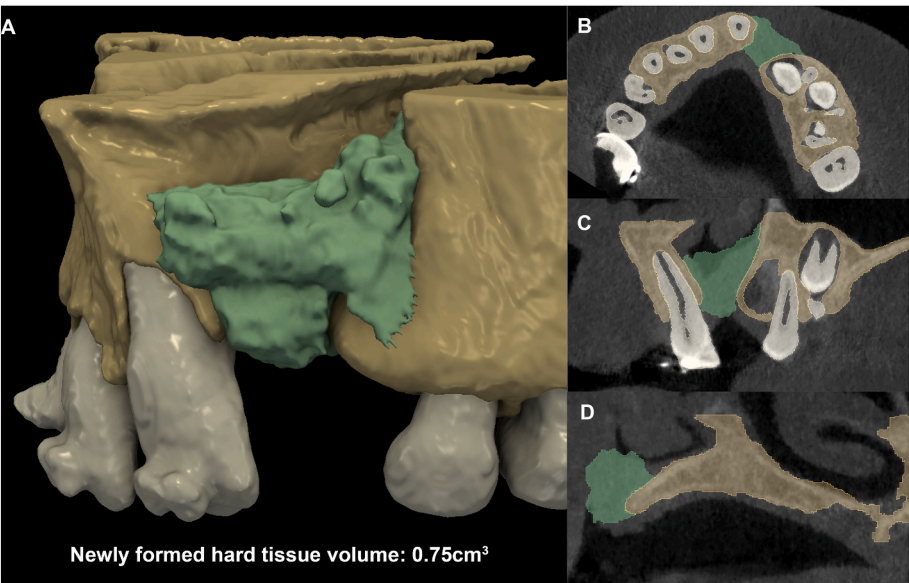


Fig. 9. Volumetric evaluation of the newly formed hard tissues: (A) 3D view; (B) axial view; (C) coronal view; (D) sagittal view.

Table 2
Planned and achieved volumetric hard-tissue gain.

Patient	Planned hard-tissue gain (cm ³)	Hard-tissue gain at 3-month follow-up (cm ³)
1	0.82	0.48
2	1.65	1.15
3	1.03	0.35
4	0.61	0.66
5	1.44	0.70
6	1.26	0.75
7	1.17	0.49
Mean ± standard deviation	1.14 ± 0.36	0.65 ± 0.26

4. Discussion

In this study the efficacy of a 3D-planned sticky ATB graft in conjunction with a split-thickness papilla curtain flap was evaluated for secondary alveolar cleft grafting. Although there have been previous case reports describing the use of ATB in secondary alveolar bone grafting (Hara et al., 2013; Jeong et al., 2015; Datarkar and Bhawalkar, 2020), these were singular case reports. To the best of our knowledge, this was the first study to report clinical and 3D-radiographic data demonstrating favorable defect fill in alveolar clefts, utilizing ATB grafting material, in a series of patients.

Although graft resorption was observed, the amount lost was in line with what can be expected from the use of iliac crest, according to the

literature (Vandersluis et al., 2020). The average volume of the remaining graft at the 3-month follow-up was well above 50% when compared with the planned graft volume. In one case, the actual graft volume even exceeded the planned volume by 8%, thereby suggesting that ATB graft could have an advantageous effect on bone regeneration.

The relatively low average DSC value (0.43 ± 0.2) and the relatively high HD ($1.83 \text{ mm} \pm 0.77 \text{ mm}$) — indicated a relatively low correlation between the preoperatively planned graft shape and the actual graft shape at 3-months follow-up. This phenomenon could be observed regardless of graft volume stability, and even cases with relatively high graft stability showed a relatively low morphological similarity to the planned graft shape. According to the 3D reconstructed images, this was due to buccal displacement of the graft during healing. In most of the cases, sufficient palatal fill could not be achieved, and the sticky ATB graft did not reach the nasal base. This was most probably due to the pressure applied on the graft by the nasal and palatal flaps; which suggests a need for further modifications to the flap design on the palatal aspect, or perhaps use of a rigid barrier on the palatal/nasal aspect in conjunction with the sticky ATB graft. A similar approach was adopted by Ishii et al. (2002), who positioned a rigid cortex bone plate on the palatal aspect to compensate for the pressure applied by the nasal/palatal flaps, and to maintain a safe, secluded space.

Limitations of the current study included the relatively low sample size, due to the pilot nature of the examination, and the lack of histological information on graft integration. Although histological evaluations have been performed previously (Radoczy-Drajko et al., 2021), only tissue integration of the ATB particles could be assessed in this

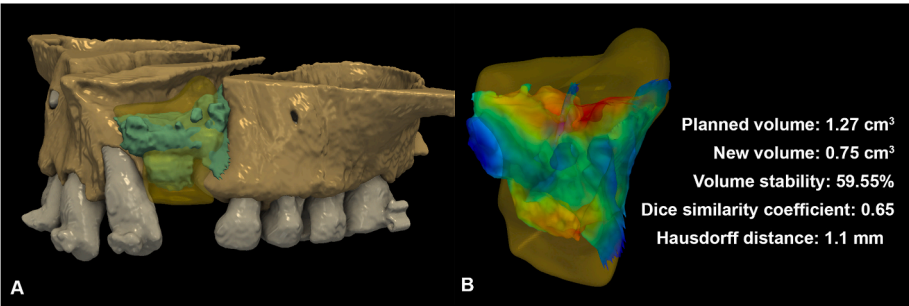


Fig. 10. Relationship between planned outcome and actual outcome: (A) 3D virtual model of planned and actual outcome; (B) colormap model visualizing the morphological differences between the newly formed hard tissues and the morphology of the planned outcome.

Table 3
Volumetric and morphological stability of the graft.

Patient	Achieved volume/ planned volume ratio (%)	Dice similarity coefficient (DSC)	Hausdorff distance (HD), average (mm)
1	58.36	0.65	0.97
2	69.27	0.57	1.56
3	34.17	0.25	2.31
4	108.21	0.41	1.91
5	48.40	0.14	3.24
6	59.55	0.65	1.10
7	41.49	0.36	1.70
Mean ± standard deviation	59.92 ± 24.35	0.43 ± 0.2	1.83 ± 0.77

current study. With the aid of 3D technology, quantitative radiographic evaluation was possible; however, the inability to perform qualitative, histological evaluation was a significant limitation.

The Bonmaker® device is suitable for preparing extracted teeth and obtaining ATB particulate grafting material. As confirmed by previous radiographic and histological studies (Radoczy-Drajko et al., 2021), ATB powder may act as a resorbable scaffold and a space-maintaining device to facilitate the healing of alveolar cleft defects. After the preparation of the ATB powder and its mixture with fibrin sealant, a sticky ATB graft was acquired with excellent intraoperative handling properties.

The time required for the harvesting of a hip graft is comparable to ATB preparation. Nevertheless, the ATB protocol has two benefits: (i) approximately half of the duration — 20 min — is fully automatic, executed by the Bonmaker device, which does not require user interaction, thus allowing medical personnel to undertake other activities during the surgery; and (ii) ATB preparation is straightforward — it does not require a highly qualified physician, with nurses fully capable of handling the graft preparation.

The estimated costs for the presented surgical approach were as follows: 3D printing 5 EUR, Bonreagent 35 EUR, fibrin glue 100 EUR. The use of these auxiliary techniques increased the costs for the operation itself, although these expenses were not considered to be exceptionally high. On the other hand, avoidance of an extraoral donor site led to reduced hospitalization time, with all patients in our study discharged from hospital on the first postoperative day. This was in contrast with the study by Brudnicki et al. (2017), who stated: ‘The mean length of hospitalization postoperatively was 2.9 days and varied from 1 to 8 days. The majority of patients were discharged on the second (66 cases) or third day (102 cases) following the procedure.’ In higher-income countries, where the cost of hospital stay is more expensive, the use of these auxiliary techniques may lower the overall costs of the treatment, but exact calculations to support this statement were beyond the scope of our study.

To ensure the compaction and tight fit of the grafting material inside the cleft, 3D preoperative planning was successfully utilized to manufacture a 3D-printed grafting template to shape the sticky ATB graft before insertion. Closure of the oronasal fistula was achieved in all cases.

Modifications of the original four-flap design, to improve certain aspects of the technique — for example, minimizing injury to the dental papillae and periodontium of the involved dentition, or increasing the amount of attached gingiva surrounding the canine — have been described previously (López-Cedrún et al., 2014). Most of these modifications continue to use a full-thickness mucoperiosteal flap, and the suture line remains directly above the graft. The novel split-thickness papilla curtain flap aims to ensure the following: (i) tension-free wound closure without compromising flap circulation, via vertical or horizontal releasing incisions; (ii) that a sufficient amount of keratinized gingiva is rotated mesially to surround the tooth on the mesial side of the cleft (usually the lateral or mesial incisor). It also has the benefit of shifting the suture line from directly above the grafted area to a more mesial position, where the sutures will rest on the well vascularized

periosteal surface of the premaxilla. The nasal and palatal mucosa on the mesial side of the cleft is still elevated in full thickness, as previously described and now widely applied.

The presented flap design resulted in primary-intention wound healing in 100% of the cases, with minimal pain and edema, as well as excellent, inflammation-free, early wound healing. These pilot results indicated the applicability of the split-thickness papilla curtain flap and, in our experience, this modification showed superior results compared with the traditional mucoperiosteal flap approach. Nevertheless, this needs to be further investigated in future comparative clinical studies.

The application of ATB as an autogenous grafting material provides a cost- and time-effective alternative for patients. Its application with the split-thickness papilla curtain flap proved to be an effective surgical approach with a low duration, while providing favorable hard- and soft-tissue conditions. This report introduces the early results of a pilot study regarding the use of ATB in alveolar cleft reconstruction, combined with a split-thickness papilla curtain flap. Although the results were favorable, the study had its limitations — namely the short follow-up period and relatively small sample size. To assess long-term outcomes of this approach each patient will need to undergo a strict follow-up protocol, with periodontal charting and follow-up intraoral scans performed on a 6-monthly basis for 5 years. CBCT scans will be performed 1 year postoperatively in order to measure long-term graft stability.

5. Conclusion

Within the limitations of the study it seems that ATB graft may be a safe and predictable grafting material in alveolar cleft reconstruction. The applied split-thickness papilla curtain flap was effectively utilized to avoid postoperative graft exposure and to improve soft-tissue conditions around the teeth adjacent to the alveolar cleft. Uneventful wound healing and favorable radiographic outcomes indicated hard- and soft-tissue stability in the grafted area. However, certain aspects of this surgical approach could be further improved, and, therefore, further research is necessary to exploit the full potential of ATB and the novel flap design in alveolar cleft grafting.

Funding

The current study received no funding.

Ethical approval

The study was approved by the university’s Regional and Institutional Committee of Science and Research Ethics (approval number: SE RKEB 251/2020). The study protocol was submitted and approved by the US National Library of Medicine (www.clinicaltrials.gov; trial registration number: NCT05971914). The research plan was compiled in line with current legislation and the Declaration of Helsinki of the World Medical Association (reference number: 23/2002. V.9.).

Patient consent

Surgical interventions were undertaken with the understanding and written consent of each subject’s caregiver.

Declaration of competing interests

The authors declare no competing interests.

Acknowledgements

Not applicable.

References

- Abyholm, F.E., Bergland, O., Semb, G., 1981. Secondary bone grafting of alveolar clefts. A surgical/orthodontic treatment enabling a non-prosthetic rehabilitation in cleft lip and palate patients. *Scand. J. Plast. Reconstr. Surg.* 15, 127–140.
- Alonso, N., Tanikawa, D.Y., Freitas Rda, S., Canan Jr., L., Ozawa, T.O., Rocha, D.L., 2010. Evaluation of maxillary alveolar reconstruction using a resorbable collagen sponge with recombinant human bone morphogenetic protein-2 in cleft lip and palate patients. *Tissue Eng. C Methods* 16, 1183–1189.
- Aly, L.A., Hammouda, N., 2016. Secondary closure of alveolar cleft with resorbable collagen membrane and a combination of intraoral autogenous bone graft and deproteinized anorganic bovine bone. *Ann Maxillofac Surg* 6, 165–171.
- Blume, O., Back, M., Born, T., Donkiewicz, P., 2019. Reconstruction of a unilateral alveolar cleft using a customized allogenic bone block and subsequent dental implant placement in an adult patient. *J. Oral Maxillofac. Surg.* 77, 2127 e2121–2127.e2111.
- Boyne, P.J., Sands, N.R., 1972. Secondary bone grafting of residual alveolar and palatal clefts. *J. Oral Surg.* 30, 87–92.
- Brudnicki, A., Sawicka, E., Brudnicka, R., Fudalej, P.S., 2017. Cephalometric comparison of early and late secondary bone grafting in the treatment of patients suffering from unilateral cleft lip and palate. *J. Cranio-Maxillo-Fac. Surg.* 45 (4), 479–484.
- Datarkar, A., Bhawalkar, A., 2020. Utility of tooth as an autogenous graft material in the defects of alveolar cleft — a novel case report. *J. Oral Biol. Craniofac Res* 10, 470–473.
- Dimitriou, R., Mataliotakis, G.I., Angoules, A.G., Kanakaris, N.K., Giannoudis, P.V., 2011. Complications following autologous bone graft harvesting from the iliac crest and using the RIA: a systematic review. *Injury* 42 (Suppl. 2), S3–15.
- Ellapakurthi, P., Reddy, G.S.P., 2021. The effectiveness of mineralized plasmatic matrix in the closure of alveolar clefts with volumetric assessment. *Regen Med Res* 9, 1.
- Eskeland, G., Borchgrevink, H., Abyholm, F.E., 1979. Columella lengthening in bilateral cleft lip patients. Experience with the forked flap procedure. *Scand. J. Plast. Reconstr. Surg.* 13, 429–436.
- Eufinger, H., Leppänen, H., 2000. Iliac crest donor site morbidity following open and closed methods of bone harvest for alveolar cleft osteoplasty. *J. Cranio-Maxillo-Fac. Surg.* 28 (1), 31–38.
- Fabian, Z., Kádár, K., Patonay, L., Nagy, K., 2019. Application of 3D printed biocompatible plastic surgical template for the reconstruction of a nasaloalveolar cleft with preoperative volume analysis. *Mater. Plast.* 56, 413–415.
- Hara, S., Mitsugi, M., Kanno, T., Tatemoto, Y., 2013. Bone transport and bone graft using auto-tooth bone for alveolar cleft repair. *J. Craniofac. Surg.* 24, e65–e68.
- Huynh-Ba, G., Brägger, U., Zwahlen, M., Lang, N.P., Salvi, G.E., 2009. Periodontal disease progression in subjects with orofacial clefts over a 25-year follow-up period. *J. Clin. Periodontol.* 36, 836–842.
- Ishii, M., Ishii, Y., Moriyama, T., Gunji, A., Morita, K., Imaizumi, F., Enomoto, S., 2002. Simultaneous cortex bone plate graft with particulate marrow and cancellous bone for reliable closure of palatal fistulae associated with cleft deformities. *Cleft Palate Craniofac J* 39, 364–369.
- Jacobs, R., Salmon, B., Codari, M., Hassan, B., Bornstein, M.M., 2018. Cone beam computed tomography in implant dentistry: recommendations for clinical use. *BMC Oral Health* 18, 88.
- Janssen, N.G., Weijjs, W.L., Koole, R., Rosenberg, A.J., Meijer, G.J., 2014. Tissue engineering strategies for alveolar cleft reconstruction: a systematic review of the literature. *Clin. Oral Invest.* 18, 219–226.
- Janssen, N.G., Schreurs, R., Gkp, Bittermann, Borstlap, W.A., Koole, R., Meijer, G.J., Tjj, Maal, 2017. A novel semi-automatic segmentation protocol for volumetric assessment of alveolar cleft grafting procedures. *J. Cranio-Maxillo-Fac. Surg.* 45 (5), 685–689.
- Jeong, K.I., Lee, J., Um, I.W., Kim, Y.K., 2015. Alveolar cleft restoration using autogenous tooth bone graft material for implant placement: a case report. *J. Oral Implantol.* 41, 487–490.
- Kadry, W., Eldeftar, M., Nassar, Y., Abou-El-Fetouh, A., Hakam, M.M., 2021. Clinical, volumetric and densitometric evaluation of tissue engineered constructs for secondary alveolar cleft reconstruction: a randomized clinical trial. *J. Cranio-Maxillo-Fac. Surg.* 49 (12), 1141–1150.
- Kamperos, G., Theologie-Lygidakis, N., Tsiklakis, K., Iatrou, I., 2020. A novel success scale for evaluating alveolar cleft repair using cone-beam computed tomography. *J. Cranio-Maxillo-Fac. Surg.* 48 (4), 391–398.
- Kim, S.Y., Kim, Y.K., Kim, H.S., Yun, P.Y., Kim, S.G., Choi, Y.H., 2017. Extraction socket sealing using palatal gingival grafts and resorbable collagen membranes. *Maxillofac. Plast. Reconstr. Surg.* 39, 39.
- Klein, S., Staring, M., Murphy, K., Viergever, M.A., Pluim, J.P., 2010. Elastix: a toolbox for intensity-based medical image registration. *IEEE Trans. Med. Imag.* 29, 196–205.
- Larson, O., Nordin, K.E., Nylén, B., Eklund, G., 1983. Early bone grafting in complete cleft lip and palate cases following maxillofacial orthopedics. II. The soft tissue development from seven to thirteen years of age. *Scand. J. Plast. Reconstr. Surg.* 17, 51–62.
- Lavagen, N., Nokovitch, L., Algrin, A., Dakpe, S., Testelin, S., Devauchelle, B., Gbaguidi, C., 2021. Efficiency of advanced-PRF usage in the treatment of alveolar cleft with iliac bone graft: a retrospective study. *J. Cranio-Maxillo-Fac. Surg.* 49 (10), 923–928.
- Linderup, B.W., Küseler, A., Jensen, J., Cattaneo, P.M., 2015. A novel semiautomatic technique for volumetric assessment of the alveolar bone defect using cone beam computed tomography. *Cleft Palate Craniofac J* 52, e47–e55.
- López-Cedrón, J.L., Gonzalez-Landa, G., Figueroa, A., 2014. Isolated keratinized gingiva incision in alveolar cleft bone grafts improves qualitative outcomes: a single surgeon's 23 year experience. *J. Cranio-Maxillo-Fac. Surg.* 42 (8), 1692–1697.
- Palankar, V., Sattur, A., Palankar, A., Rajeswari, S.R., Thakur, S., Desai, A.K., 2021. Evaluation of long-term stability of secondary alveolar bone grafts in cleft palate patients using multislice computed tomography and three-dimensional printed models: a prospective study. *J. Pharm. BioAllied Sci.* 13, S1496–S1500.
- Pálházi, P., Nemes, B., Swennen, G., Nagy, K., 2014. Three-dimensional simulation of the nasaloalveolar cleft defect. *Cleft Palate Craniofac J* 51, 593–596.
- Palkovics, D., Bolya-Orosz, F., Pinter, C., Molnar, B., Windisch, P., 2022a. Reconstruction of vertical alveolar ridge deficiencies utilizing a high-density polytetrafluoroethylene membrane/clinical impact of flap dehiscence on treatment outcomes: case series. *BMC Oral Health* 22, 490.
- Palkovics, D., Molnar, B., Pinter, C., Gera, I., Windisch, P., 2022b. Utilizing a novel radiographic image segmentation method for the assessment of periodontal healing following regenerative surgical treatment. *Quintessence Int.* 53, 492–501.
- Palkovics, D., Pinter, C., Bartha, F., Molnar, B., Windisch, P., 2021a. CBCT subtraction analysis of 3D changes following alveolar ridge preservation: a case series of 10 patients with 6-months follow-up. *Int. J. Comput. Dent.* 24, 241–251.
- Palkovics, D., Solyom, E., Molnar, B., Pinter, C., Windisch, P., 2021b. Digital hybrid model preparation for virtual planning of reconstructive dentoalveolar surgical procedures. *J. Vis. Exp.* (174).
- Pinter, C., Lasso, A., Fichtinger, G., 2019. Polymorph segmentation representation for medical image computing. *Comput. Methods Progr. Biomed.* 171, 19–26.
- Radoczy-Drajko, Z., Windisch, P., Svidro, E., Tajti, P., Molnar, B., Gerber, G., 2021. Clinical, radiographical and histological evaluation of alveolar ridge preservation with an autogenous tooth derived particulate graft in EDS class 3–4 defects. *BMC Oral Health* 21, 63.
- Sahni, V., Grover, V., Sood, S., Jain, A., 2022. The periodontal status of orofacial cleft patients: a systematic review and meta-analysis. *Cleft Palate Craniofac J*, 10556656221127549.
- Segura-Castillo, J.L., Aguirre-Camacho, H., González-Ojeda, A., Michel-Perez, J., 2005. Reduction of bone resorption by the application of fibrin glue in the reconstruction of the alveolar cleft. *J. Craniofac. Surg.* 16, 105–112.
- Stasiak, M., Wojtaszek-Słomińska, A., Racka-Pilszka, B., 2019. Current methods for secondary alveolar bone grafting assessment in cleft lip and palate patients — a systematic review. *J. Cranio-Maxillo-Fac. Surg.* 47 (4), 578–585.
- Vandersluijs, Y.R., Fisher, D.M., Stevens, K., Tompson, B.D., Lou, W., Suri, S., 2020. Comparison of dental outcomes in patients with nonsyndromic complete unilateral cleft lip and palate who receive secondary alveolar bone grafting before or after emergence of the permanent maxillary canine. *Am. J. Orthod. Dentofacial Orthop.* 157, 668–679.
- Wu, C., Pan, W., Feng, C., Su, Z., Duan, Z., Zheng, Q., Hua, C., Li, C., 2018. Grafting materials for alveolar cleft reconstruction: a systematic review and best-evidence synthesis. *Int. J. Oral Maxillofac. Surg.* 47, 345–356.
- Yuan, K.F., Lai, Q.G., Qi, C., Guo, X.H., Shi, R.J., Xu, X., Wei, F.C., 2004. [Clinical study of bioglass for repairing alveolar cleft. *Shang Hai Kou Qiang Yi Xue* 13, 465–468.